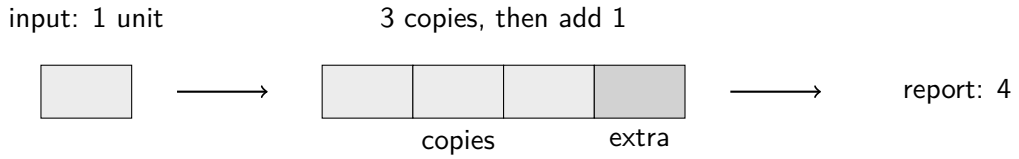
For Problems 1–2, a machine makes 1, 2, or 3 copies of the input length, then adds 0, 1, 2, 3, 4, 5, or 6 units. It keeps both choices fixed for the whole round. Each square in the paper row is one unit.



Problem 1: Play the machine. Secretly choose the number of copies and the extra amount. Your partner requests a strip's report. They may request more reports until they can name both choices. Reveal and switch jobs. How few requests can you manage?

Round	Strips requested	Reports
1		
2		
3		
4		



A

B

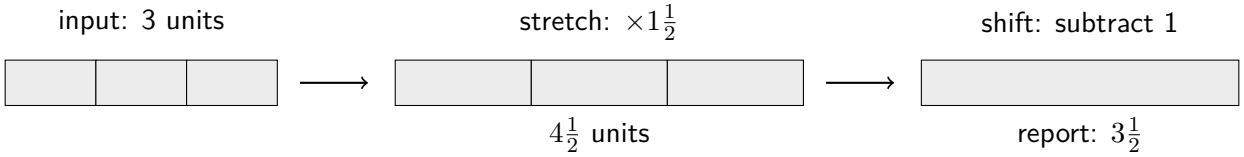
C

D

Problem 2: A 2-unit strip reports 6. Find every possible machine. Choose a second strip that would separate all your possibilities, and show the reports you might get. Could a single first request ever identify a machine?

Copies	Extra amount	Second strip and its report

For the remaining problems, a stretch-and-shift machine multiplies every input length by the same positive number, then adds the same fixed amount. Fractions are allowed; the added amount may be negative.



Problem 3: Each machine gives these two reports. Find its rule and its reports for lengths 0, 4, and 10. Could a different stretch-and-shift rule give the same two known reports?

Machine A	Machine B
2 units $\to$ 8 6 units $\to$ 16	2 units $\to$ 8 6 units $\to$ 12
Rule:	Rule:
0 units $\to$	0 units $\to$
4 units $\to$	4 units $\to$
10 units $\to$	10 units $\to$

Problem 4: A machine claims to use one fixed stretch and shift. Could these three reports be correct? Find every way to repair the list by changing exactly one report.

2 units $\to$ 7	4 units $\to$ 11	6 units $\to$ 16

Problem 5: A machine gives 7, 11, and 15 for lengths 2, 4, and 6. Its rule stays fixed, but nobody promises stretch-and-shift. Make two complete report tables that fit these reports and disagree at length 8. Could any number of matching tests below length 8 force its report at 8?

Length	1	2	3	4	5	6	7	8
Table A		7		11		15		
Table B		7		11		15		